Node.js

Ch 1: introduction and installation

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D. HOSTENS & M. DIMA

— Lecturers

Dirk Hostens (dirk.hostens@vives.be)

Milan Dima (milan.dima@vives.be)

— Ground Rules

- Grading: 100% Project + oral defense examen

  See Toledo for details (Practical organization)

- Questions:

  - (practical + course material questions) via Forum on Toledo!

  - Private matters => mail /Teams

- Best preparation => presence + exercises

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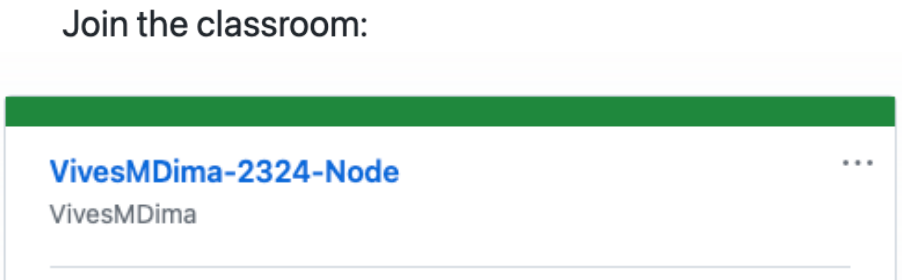
— Git (recap)

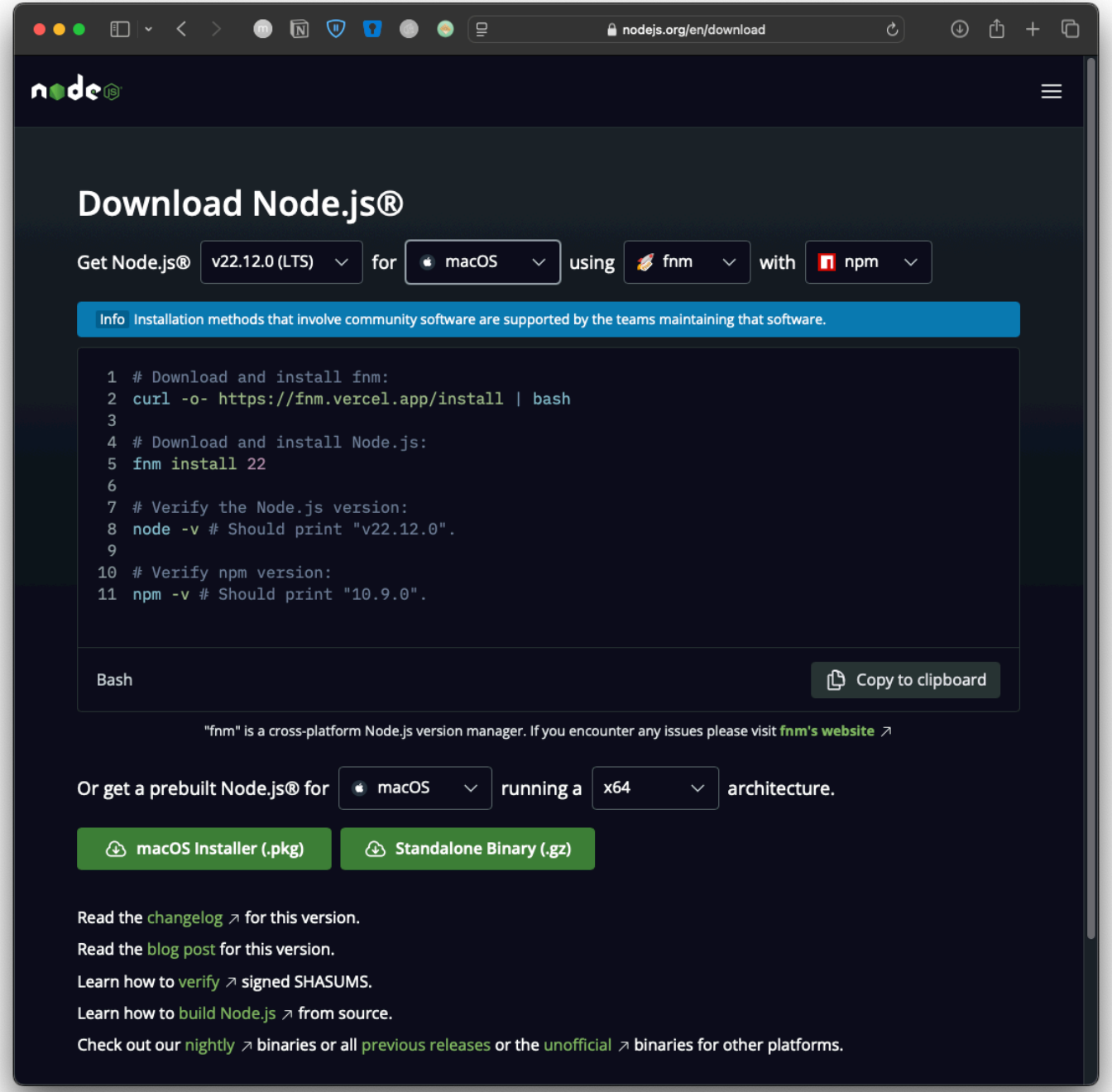


```
git init
git remote add origin [URL]
git add .
git commit - m ‘eerste commit’
git push origin main (master?)
```

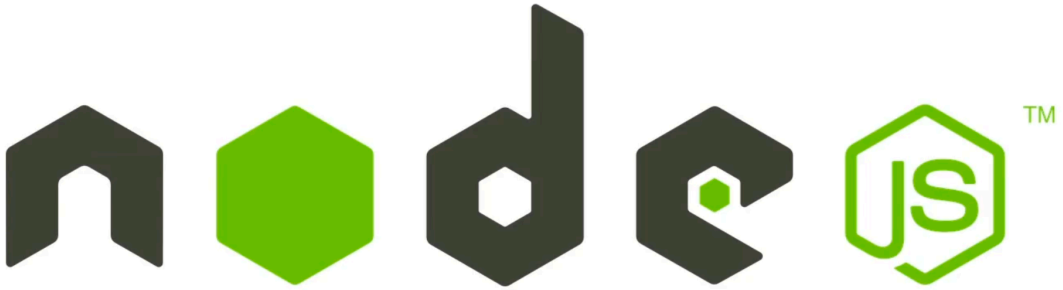


Github classroom




nodejs.org/en/download
Download Node.js®
Get Node.js® v22.12.0 (LTS) for macOS using fnm with npm
Info Installation methods that involve community software are supported by the teams maintaining that software.
1 # Download and install fnm:
2 curl -o- https://fnm.vercel.app/install | bash
3
4 # Download and install Node.js:
5 fnm install 22
6
7 # Verify the Node.js version:
8 node -v # Should print "v22.12.0".
9
10 # Verify npm version:
11 npm -v # Should print "10.9.0".
Bash
Copy to clipboard
"fnm" is a cross-platform Node.js version manager. If you encounter any issues please visit fnm's website ↗
Or get a prebuilt Node.js® for macOS running a x64 architecture.
macOS Installer (.pkg)
Standalone Binary (.gz)
Read the changelog ↗ for this version.
Read the blog post for this version.
Learn how to verify ↗ signed SHASUMS.
Learn how to build Node.js ↗ from source.
Check out our nightly ↗ binaries or all previous releases or the unofficial ↗ binaries for other platforms.

What is Node?



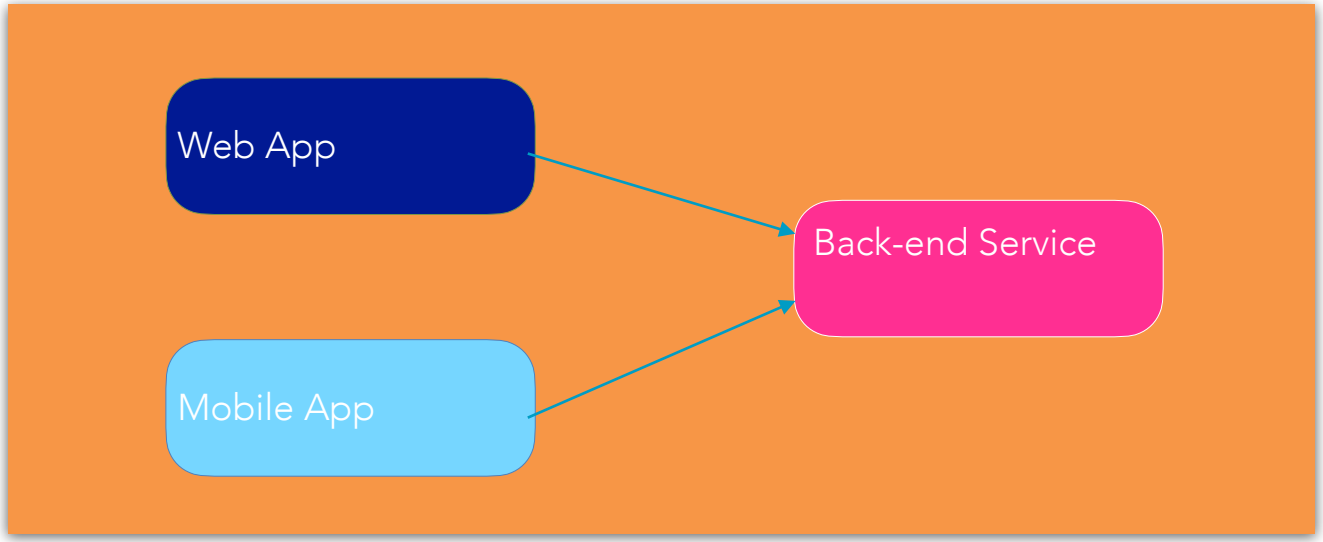
- Open Source
- Cross-platform
- Runtime environment
- To run javascript (or TS) outside the browser

A **runtime environment** for executing JavaScript code

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What is it used for?

- Usually for Back-end Services
- Like API's (Application Programming Interface)
- Services powering Client Applicaties



— Ideal for:

Highly-scalable, data-intensive and real-time apps

Great for prototyping and agile development

Superfast and highly scalable

- Used in Production by big companies like:
- Paypal, Netflix, Uber, LinkedIn, NASA, eBay, Groupon, Mozilla, Aliexpress…

Source: https://trio.dev/blog/companies-use-node-js

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— Used by:

15+ Popular Companies Using Node.js in 2024

Here are the companies using Node.js: Netflix, NASA, Trello, PayPal, LinkedIn, Walmart, Uber, Twitte Yahoo, eBay, GoDaddy, and got much better results.

Tejas Kaneriya
March 3, 2021

15 mins read Last Updated February 02, 2

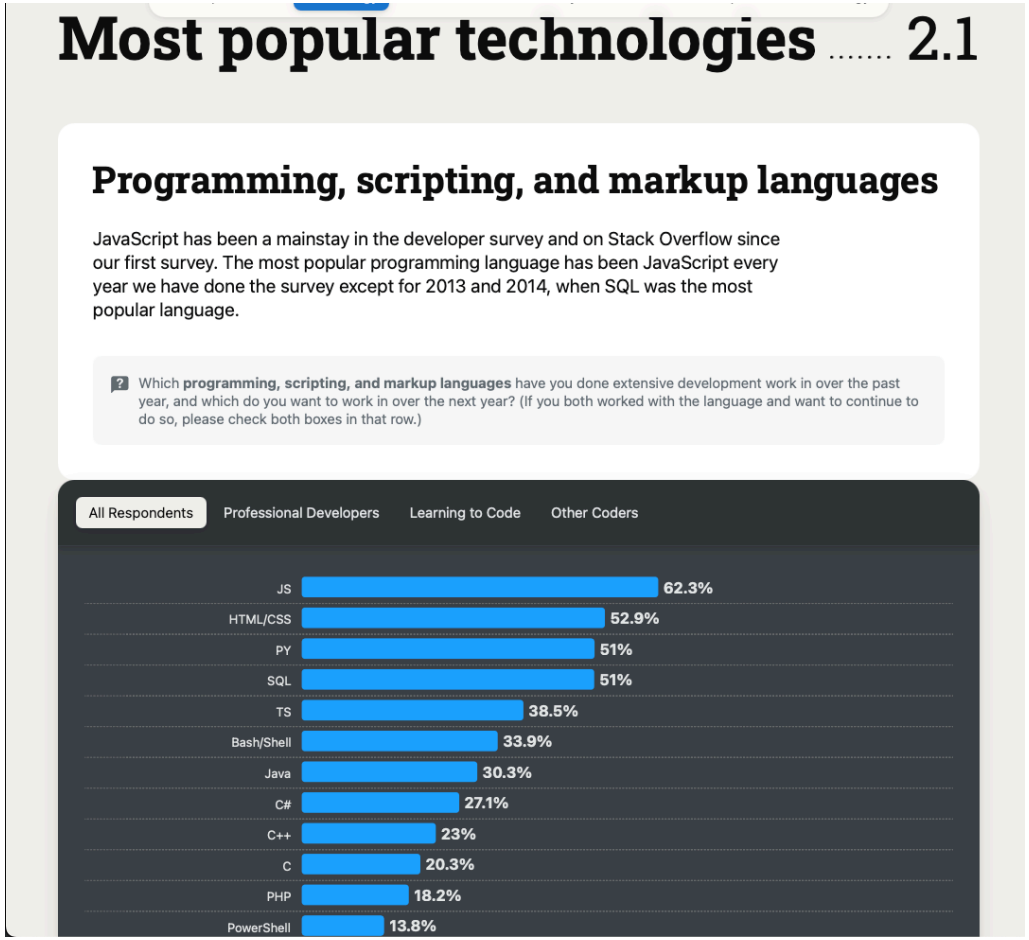


Bron: https://www.simform.com/blog/companies-using-nodejs/

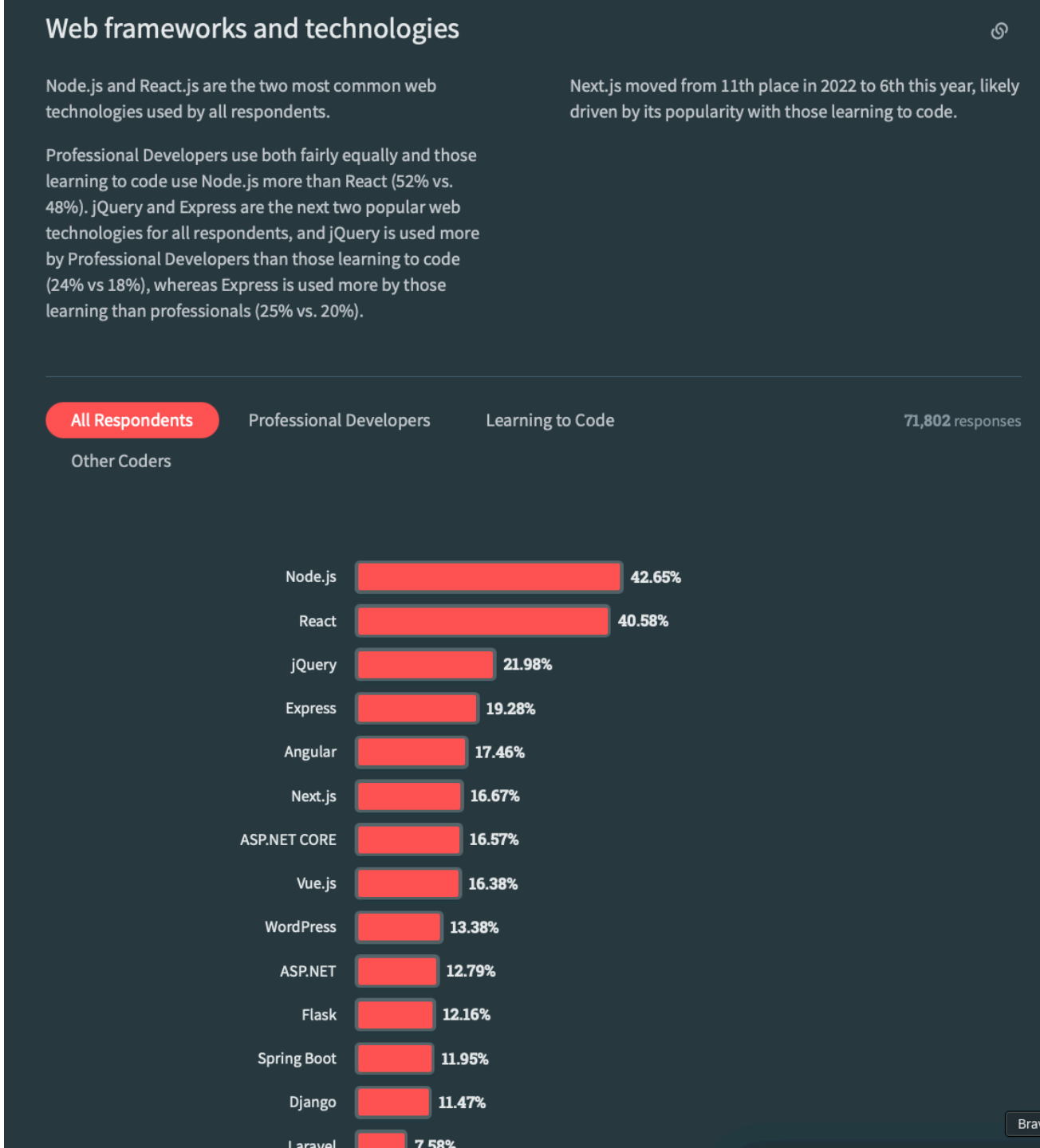
1. Netflix lowers startup time with Node.js.
2. NASA improved database access time with Node.js.
3. Trello achieved quick prototyping with Node.js.
4. PayPal decreased its loading time with Node.js.
5. Linkedin improved its app performance with Node.js.
6. Walmart improved website experience with Node.js
7. Uber eases ride processing with Node.js.
8. Twitter achieves 5 seconds loading speed with Node.js.
9. Yahoo simplified feature releases with Node.js.
10. eBay reduced resource utilization with Node.js.
11. GoDaddy showed 4x higher end-user performance with Node.js.
12. Groupon achieved scalability with Node.js.
13. Medium speeds up content processing with Node.js.
14. Yandex created numerous APIs with Node.js.
15. Citibank achieved 50% lower latency with Node.js.

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— StackOverflow Developer Suvery 2024

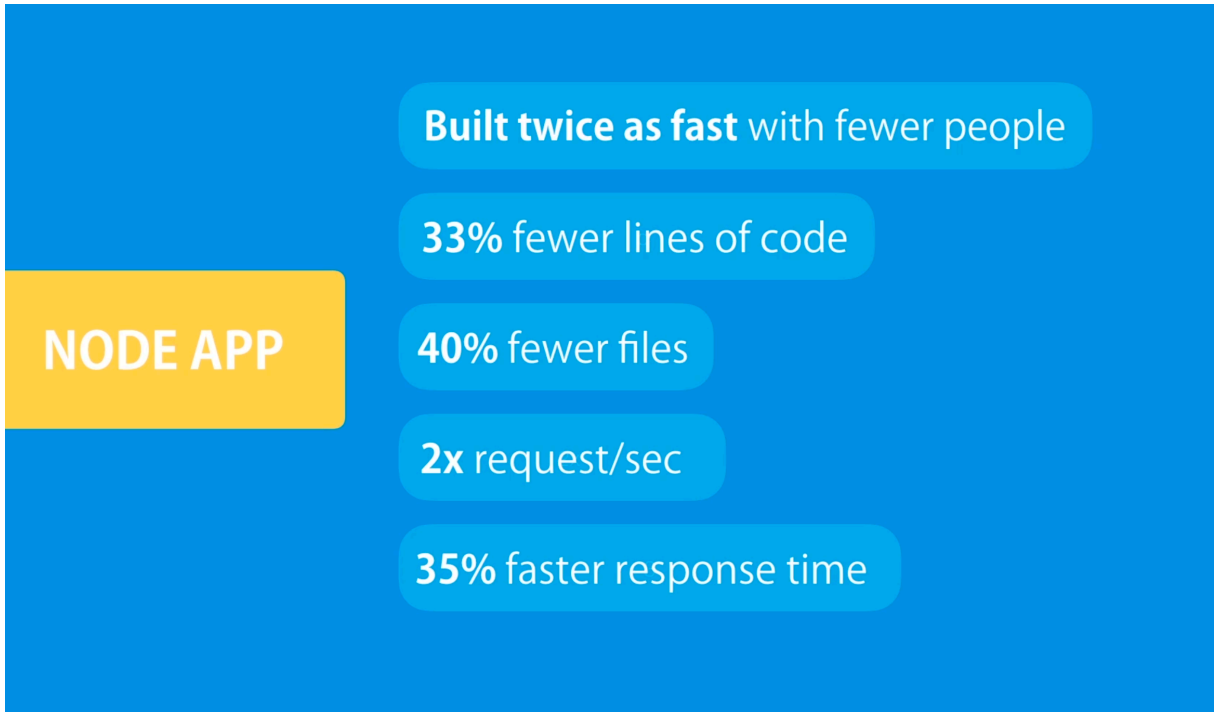


https://survey.stackoverflow.co/2023/#most-popular-technologies-webframe



— Case Paypal

Java Spring Application -> Node.js



Bron: https://engineerbabu.com/blog/node-js-the-rising-technology-behind-paypal/

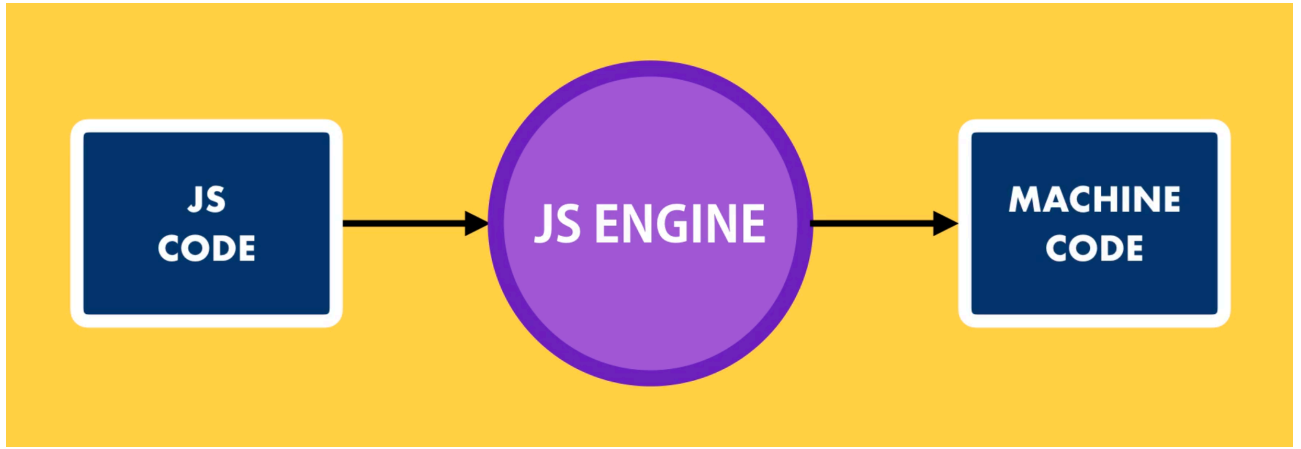
— Key points

- Great for prototyping and agile development
- Super fast and highly scalable
- JavaScript everywhere
- Cleaner and more consistent code
- Large ecosystem of open-source libs

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— Before Node:

Javascript was meant for inside the browser

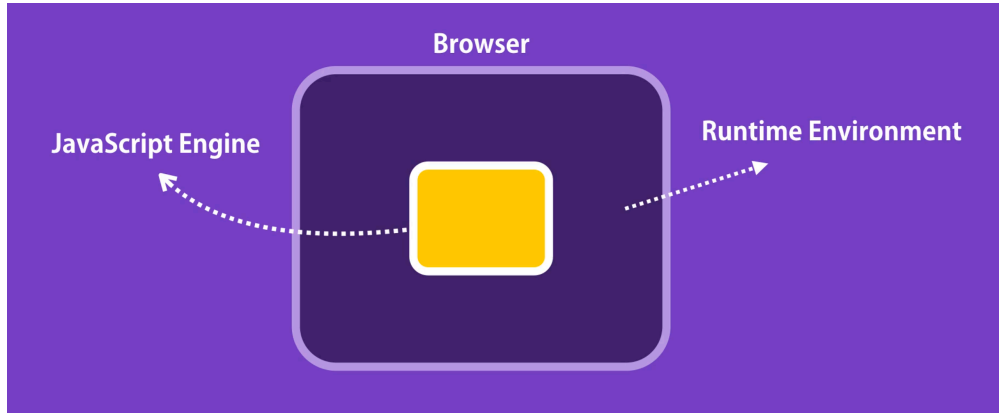


Different browsers = different engines!



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— Browser provides runtime environment



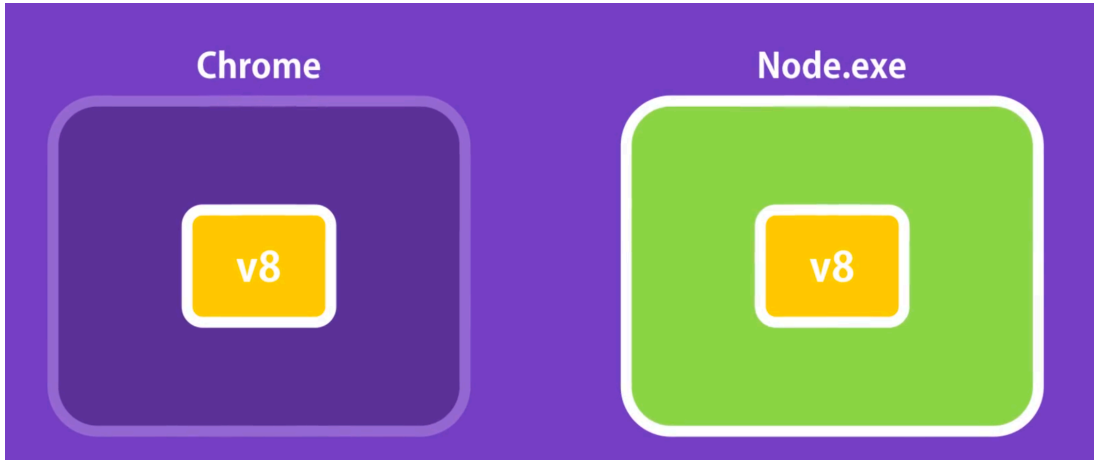
Elements like

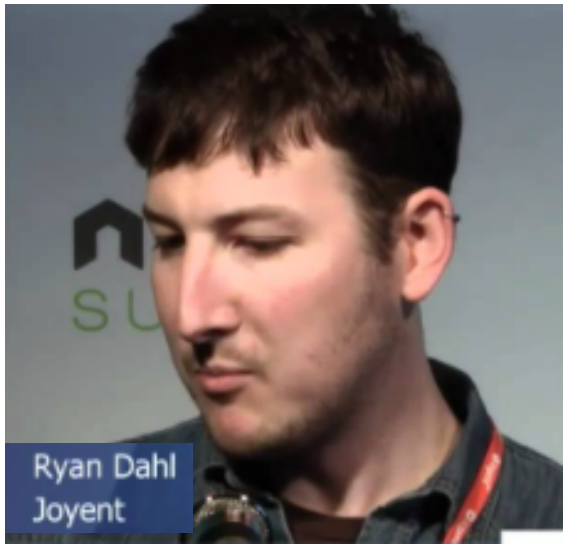
document & window

```
document.getElementById('');
```

In 2009 Ryan Dahl -> JS outside the browser!

V8 embedded in C++ program = Node





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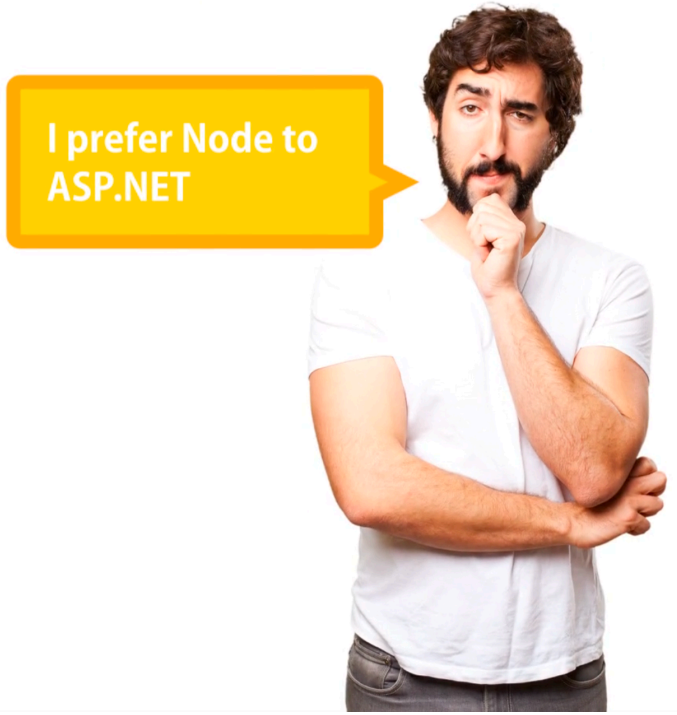
— Node

- Uses objects too
- Differs from browsers browsers

~~`document.getElementById('');`~~

```
fs.readFile()
```

```
http.createServer()
```



Recap: Node is an application containing a V8 JS engine + additional modules for extra functionality. (File System, network …)

Is not a FRAMEWORK! <> ASP.net, Rails, Django

Node is NOT a programming language!

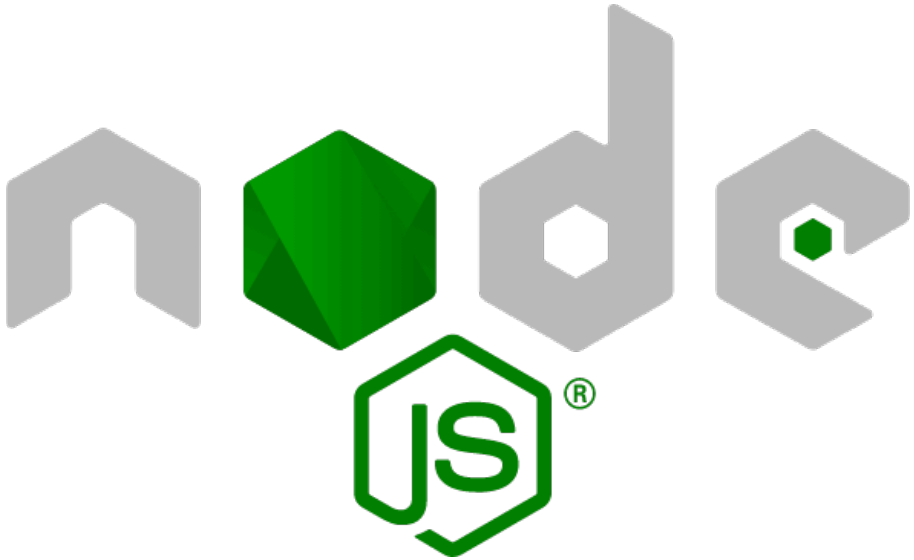
Node is NOT a framework!

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— Hoe does Node work?

Node applications are

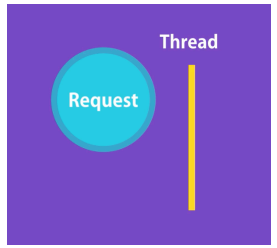
- Very scalable
- Real-time
- Non-blocking -> Asynchronous
- Javascript

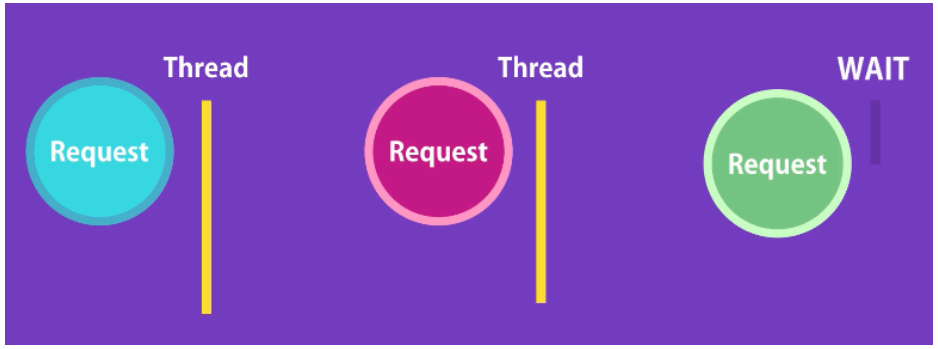


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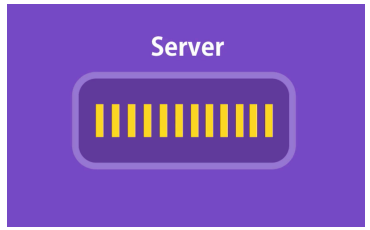
— Synchronous (usually not in Node)

- Request comes in (e.g. DB call)
- Thread is allocated and waits for end
- New request = new thread





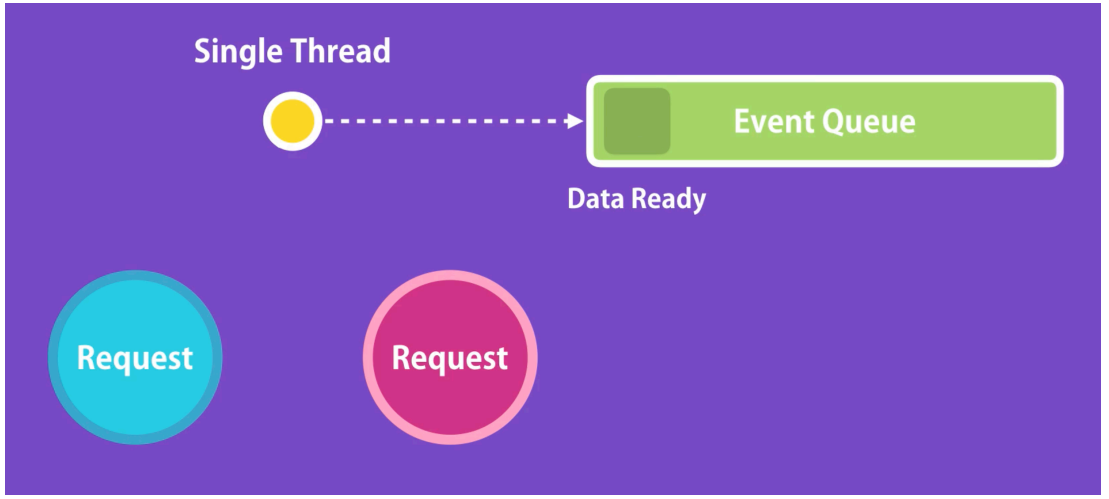
- With many clients: threads get exhausted!
- -> clients have to wait or deploy more hardware
- E.g. default ASP.net



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— With Node (asynchronous)

- Only 1 thread. Handels all requests
- Event Queue contains responses (is constantly monitored)
- E.g. read from DB -> While data is downloaded
- Thread handels next request



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— What is asynchronous?

Restaurant analogy:

- Waiter writes order by order (Asynchronous)
- Waiter writes down one order and waits until it's ready to deliver it and write the next order down. (Synchronous)
  Several waiters = threads



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— Node is good for:

Intensive I/O Applications (++ disk or network access)



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— Node is not good for:

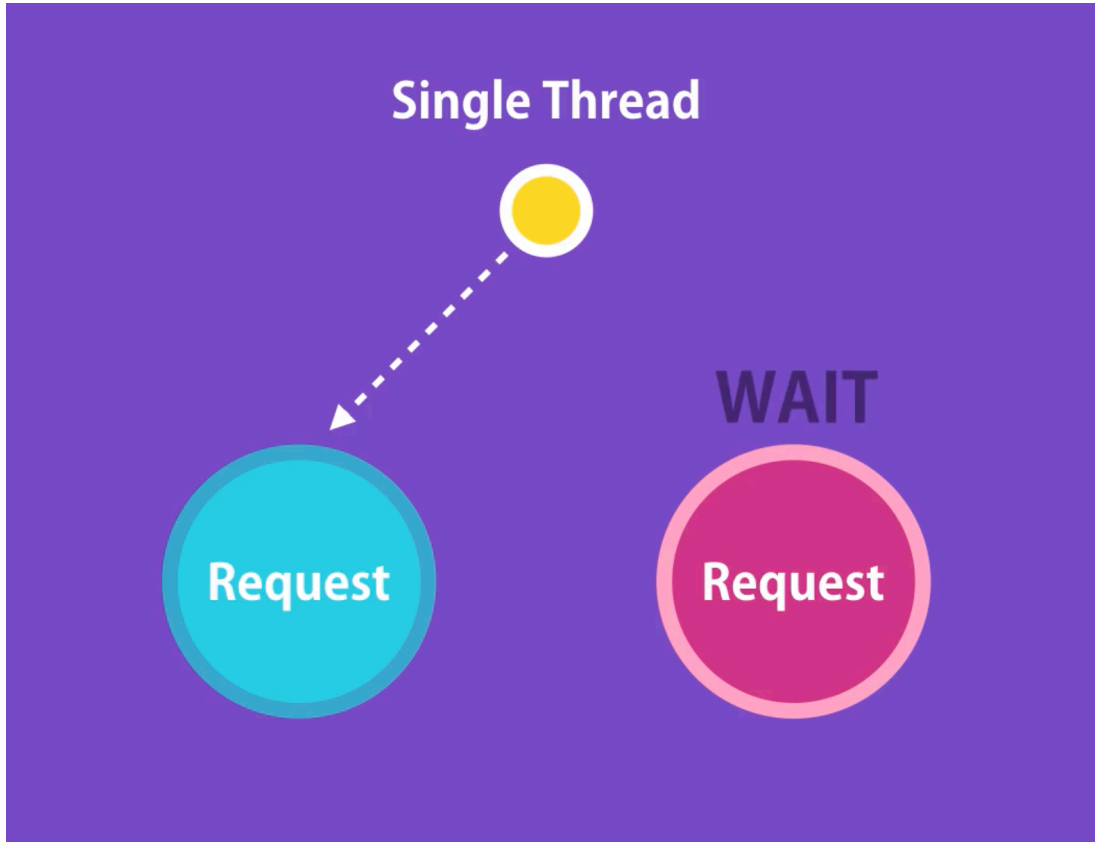
CPU Intensive Applications (Video-encoding, Photo ...)



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— Single thread with CPU intensive use

With CPU intensive operations, the other clients need to wait until the calculations are finished (CPU is free)



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— Install Node

https://nodejs.org/en/

Windows: https://nodejs.org/dist/v22.12.0/node-v22.12.0-x64.msi

MacOS 64-bit & ARM: https://nodejs.org/dist/v22.12.0/node-v22.12.0.pkg (**or** `brew install node`)

Linux: https://nodejs.org/dist/v22.12.0/node-v22.12.0-linux-x64.tar.xz (**or** `sudo apt install nodejs`)

Docker: https://hub.docker.com/_/node/

Others: https://nodejs.org/en/download/

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— IDE

Free Choice of IDE

Visual Studio Code, WebStorm, Atom, Sublime, IntelliJ Idea, Komodo, Webmatrix, Koding, Brackets ...

Slides: VS Code https://code.visualstudio.com/docs/nodejs/nodejs-tutorial

Plugins: npm intellisense, ES Lint, Prettier, REST Client (Document This)

Also see: https://docs.microsoft.com/en-us/shows/beginners-series-to-nodejs/how-to-setup-vs-code-for-nodejs-development-5-of-26

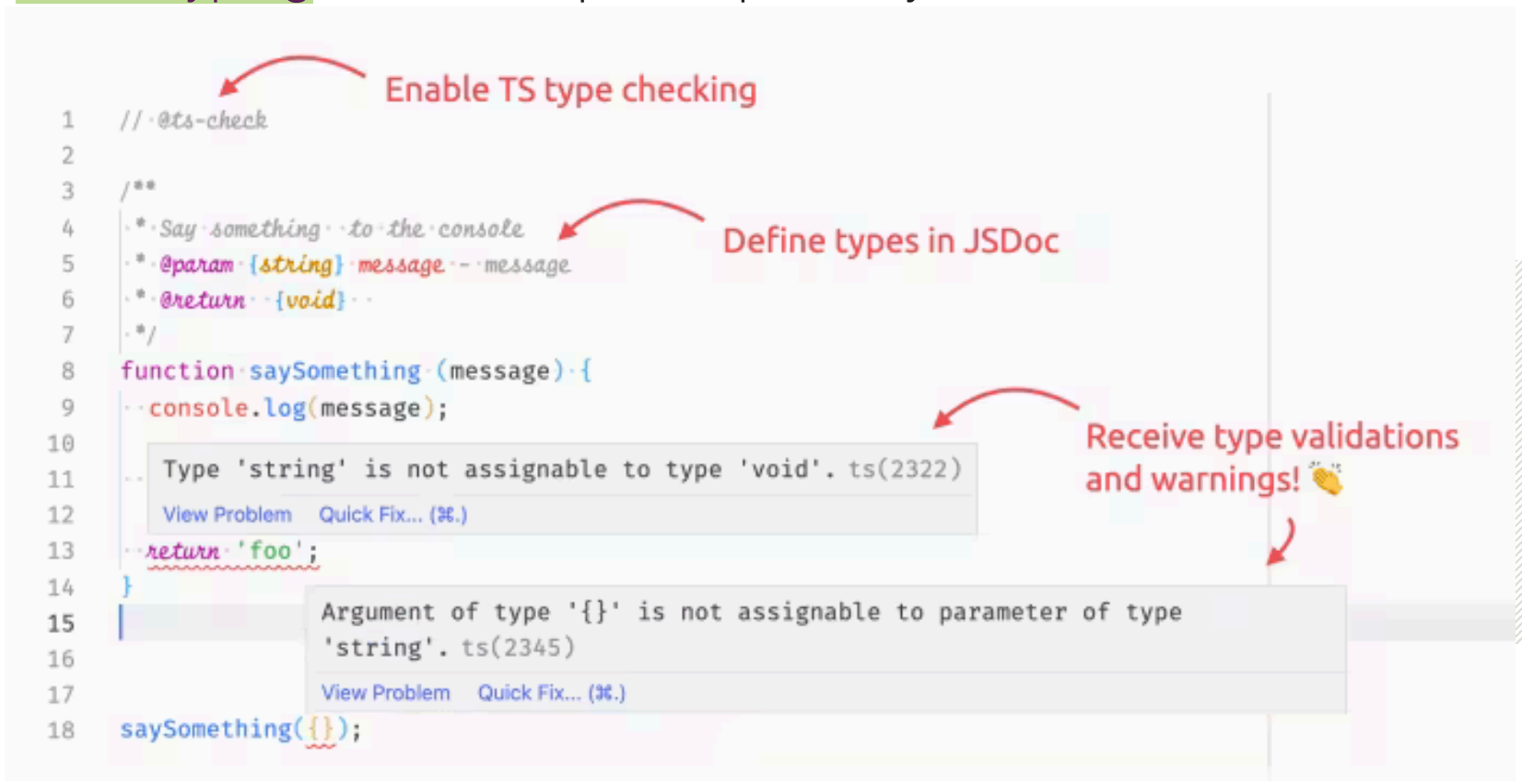
— JavaScript vs TypeScript vs JSDoc

Javascript: known from previous course web development

Typescript: Developed by MS, Static Typing, ECMAScript compatibility, own compiler. Used in Angular?

JSDoc: Markup language and documentation tool. Type checking with Annotations!

https://jsdoc.app/



— Install

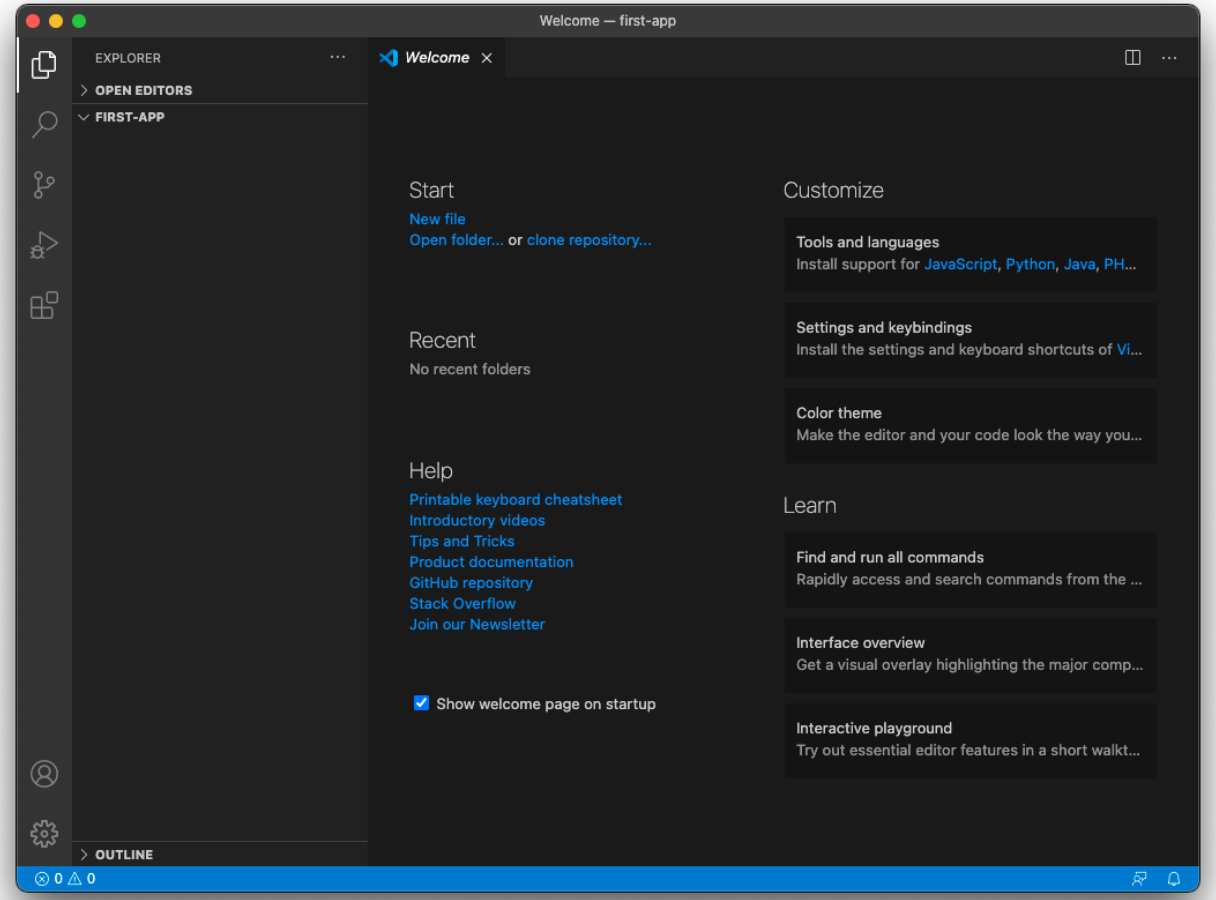
```
~ ➤ brew install node
Running `brew update --preinstall`...
...
...
~ ➤ node --version
v23.5.0
~ ➤
```

Install Latest Stable Version
Command line or installer

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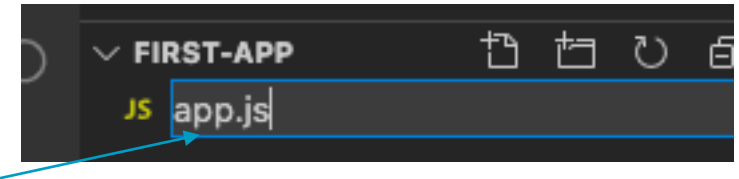
First application

```
milan@Dev〽 mkdir first-app
milan@Dev〽 cd first-app
milan@first-app〽 code .
```



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— First application (cont.)



Create a new file `app.js`

```
function sayHello(name){
    console.log('Hello ' + name );
}
sayHello('Vives');
```

Run Code:

```
milan@first-app〽 node app.js
Hello Vives
milan@first-app〽
```

Try: `console.log(window);`

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— Keys VS Code

`Ctrl +`` = open integrated terminal

`Ctrl + Shift + `` = create new terminal

`Ctrl + Shift + P` = show Command Palette

`Shift+Alt+↓ / ↑` = copy current line lijn to above/under

`Ctrl (Cmd mac)+ Shift + L` = select all instances of current selection

`Alt + Shift + A`: comment/uncomment block

`Ctrl (Cmd mac) + Shift + :` = comment/uncomment line

https://code.visualstudio.com/shortcuts/keyboard-shortcuts-windows.pdf

https://code.visualstudio.com/shortcuts/keyboard-shortcuts-macos.pdf

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— Lab:

Download the starter from Github Classroom (see Toledo for link). Write a program that divides the students in 7 different groups based on their date of birth (YYYYMMDD hardcoded) (Tip modulus-division)

Extra 1: User is asked to input his date of birth in terminal

```
milan@first-app〽 node groepen.js
Insert date of birth (YYYYMMDD): 20201010
You are assigned to group 5
milan@first-app〽
```

Extra 2: see next slide

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— Lab cont.

Extra 2: Write a switch statement where depending on picked group another Lorem Ipsum sentence is shown.

```
milan@first-app〽 node groepen-switch.js
Insert date of birth (YYYYMMDD): 20201010
You are assigned to group 5
Your quote of the day is:
Sed ut perspiciatis unde omnis iste natus error sit
voluptatem accusantium doloremque laudantium, totam rem
aperiam, eaque ipsa quae ab illo inventore veritatis et
quasi architecto beatae vitae dicta sunt explicabo.
milan@first-app〽
```


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